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RESEARCH ARTICLE

Emergency-Aware Adaptive Traffic Signal Control Using YOLO-Based Light-Bar Detection and SUMO System-Level Validation

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ABSTRACT This paper presents a real-time vision-based framework for emergency-aware adaptive traffic signal control using roof-mounted siren/light-bar detection as a common visual cue for emergency vehicles. Unlike acoustic sensing or full-vehicle classification, the proposed method detects the light-bar itself to improve robustness across heterogeneous emergency fleets. A YOLO26m object detector was trained on a curated dataset of 3,130 manually annotated images containing both activated and non-activated light-bar states, formulated as a single-class object-presence task independent of vehicle type. Across three training runs, the detector achieved mean precision, recall, mAP50, and mAP50–95 of 0.73 ± 0.03 , 0.69 ± 0.002 , 0.73 ± 0.02 , and 0.34 ± 0.01 , respectively, with the best run reaching 0.76 precision and 0.75 mAP50. The detector was integrated as a supervisory layer over a pre-optimized Genetic Algorithm–Stochastic Gradient Descent (GA–SGD) traffic controller to activate temporary emergency priority and restore optimized operation after vehicle passage. Simulation of Urban Mobility (SUMO)-based system-level validation was performed on a modeled Souissi roundabout in Rabat, Morocco, comparing Fixed-time, GA–SGD, and the proposed GA–SGD with vision-triggered priority override. In the selected stress scenario, the proposed framework reduced ambulance travel time to 56 s, compared with 70 s for GA–SGD and 64 s for Fixed-time control, while lowering ambulance waiting time from 21 s to 15 s and total network waiting time to 19,887 s. The results support the feasibility of the proposed framework for pre-deployment intelligent traffic management studies.

INDEX TERMS Adaptive traffic signal control, emergency vehicle preemption, intelligent transportation systems, light-bar detection, SUMO, vision-based detection, YOLO26m.

I. INTRODUCTION

Rapid urbanization and increasing vehicle density have made traffic congestion a major concern for modern municipal governments, with direct impacts on economic productivity and environmental sustainability. Recent advances in intelligent transportation systems (ITS) have introduced sophisticated metaheuristic-based optimization techniques to address these complexities. Among these, hybrid genetic algorithm-stochastic gradient descent (GA–SGD) has shown significant effectiveness in maximizing intersection

throughput by dynamically adjusting signal timing based on real-time vehicle distribution.

However, one important limitation remains inherent in most high-performance controllers: “emergency blindness.” While systems such as the GA–SGD approach successfully reduce average waiting times for the general population, as demonstrated in our previous study [1], they often lack the specific sensitivity required to identify and prioritize emergency vehicles (EVs). In safety-critical scenarios, we observe that ambulances, fire engines, and police vehicles experience the same optimized delays as regular traffic, potentially resulting in life-threatening response times. Additionally, these conventional preemption methods often require costly

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hardware modifications that are difficult to scale across complex urban networks.

The objective of this research is to introduce a real-time computer vision module capable of detecting the presence of an emergency light-bar (siren unit) on vehicles to enable supervisory priority control of the traffic cycle. Rather than attempting to classify entire vehicle categories or infer flashing state, the proposed approach focuses on a recurring visual cue across emergency fleets—the roof-mounted siren unit. This design choice improves generalization across regions where ambulance, police, and fire vehicles exhibit heterogeneous shapes and colors. The study explicitly builds upon the validated GA–SGD optimization framework by integrating a safety-priority module that responds to the visual existence of the siren/light-bar, ensuring that the system remains efficient during steady-state traffic while reacting with low latency to emergency presence.

In our previous work, a hybrid Genetic Algorithm–Stochastic Gradient Descent (GA–SGD) framework was developed to optimize traffic signal timing through simulation-based learning, demonstrating improved intersection efficiency in a Moroccan case study [1]. However, that framework was limited to normal traffic optimization and did not incorporate emergency vehicle detection, priority arbitration, or emergency-specific system-level validation. The present study addresses this limitation by introducing a vision-based emergency light-bar detection module that operates as a supervisory layer over the pre-optimized GA–SGD controller. In contrast to the previous work, the proposed framework is specifically designed for emergency-aware traffic signal control and is validated through SUMO-based simulations under emergency-vehicle scenarios, comparing fixed-time control, GA–SGD control, and the proposed GA–SGD with vision-triggered priority override.

The remainder of this paper is organized as follows. Section II reviews recent advances in traffic signal control, emergency vehicle preemption, deep learning in intelligent transportation systems, and YOLO-based real-time detection. Section III presents the integrated framework, combining GA–SGD traffic optimization, YOLO26m light-bar detection, and the real-time supervisory priority mechanism. Section IV describes the methodology, including dataset construction, model configuration, training and validation procedures, and priority decision logic. Section V reports the experimental results, including detection metrics and the SUMO-based case study. Finally, the last section concludes the paper, discusses limitations, and outlines future work.

II. RELATED WORK

A. RECENT ADVANCES IN TRAFFIC CONTROL AND EMERGENCY PREEMPTION

Research on emergency vehicle prioritization and intelligent traffic management has evolved along three main directions: IoT-based signal preemption, acoustic siren recognition, and vision-based vehicle detection using deep learning.

Early efforts focused primarily on IoT and connected-vehicle infrastructures. Sasikala et al. [2] proposed an IoT-enabled emergency prioritization system integrating GPS tracking, RFID identification, and adaptive signal control, reporting a 35% reduction in ambulance response time through dynamic route optimization. Zlatkovic et al. [3] investigated preemption strategies in connected vehicle environments using VISSIM simulation, demonstrating up to 36% delay reduction for emergency vehicles with minimal disruption to surrounding traffic. Similarly, Prajwal et al. [4] introduced a smart GPS-based framework where IoT devices installed at intersections pre-empt signals in real time, while Chowdhury et al. [5] extended this concept by employing UAV-assisted monitoring to balance emergency priority with the impact on adjacent traffic streams. These approaches, although effective, rely heavily on dedicated communication infrastructure and explicit vehicle identification, limiting applicability in cities where such systems are unavailable.

Another research stream has addressed acoustic siren detection. Giordano et al. [6] developed an embedded CNN-based system for real-time audio siren recognition on Raspberry Pi hardware using curated datasets and probability smoothing to reduce false positives. Jayakumar et al. [7] further improved audio classification through combined temporal and spectral features, achieving 99.5% accuracy with SVM and ensemble learning. Meiramnganba et al. [8] explored multi-sensor fusion using acoustic, seismic, and magnetic signals to enhance robustness under noisy environments. Despite promising results, purely acoustic solutions suffer from urban noise interference, reverberation, and lack of spatial localization, which restrict their reliability for intersection-level control.

Recent years have seen rapid progress in vision-based emergency vehicle recognition using YOLO architectures. Phuong et al. [9] evaluated YOLOv5–v10 models in complex Vietnamese traffic scenes, highlighting challenges with small objects and occlusion. Khalili and Smyth [10] proposed SOD-YOLOv8 to enhance small-object detection through multi-path feature fusion and PIoU loss, achieving notable gains in mAP. Fahim [11] fine-tuned YOLOv9 for vehicle detection in Dhaka traffic, while Vu [12] demonstrated retrained YOLOv8 for pedestrian monitoring at intersections. Additional studies on YOLOv11 and lightweight variants [13], [14], [15] confirmed the suitability of modern detectors for real-time ITS applications.

Specific to emergency vehicle detection, Suhana et al. [16], Asha [17], Satyanarayana et al. [18], and Sharma et al. [19] trained YOLO models to classify ambulances, fire trucks, and police cars from CCTV streams, often using vehicle appearance as the primary cue. However, these methods depend on vehicle class recognition rather than the operational indicator common to all emergency vehicles—the roof light-bar unit. Consequently, they struggle with unmarked vehicles, regional variations, and situations where class appearance is ambiguous.

The emergence of YOLO26 introduced architectural changes aimed at end-to-end inference without NMS, dynamic supervision loss, and small-target-aware label assignment [20], [21], making it suitable for detecting compact objects such as light-bars in wide intersection views. Nevertheless, existing literature has not exploited YOLO26 for this purpose, nor integrated such detection with high-level traffic optimization.

In summary, prior work either (i) relies on infrastructure-dependent IoT preemption [2], [3], [4], [5], (ii) employs acoustic detection with limited spatial reliability [6], [7], [8], or (iii) performs vehicle-class recognition rather than light-bar presence detection [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19]. A gap remains for a non-intrusive, vision-based module focused solely on the light-bar as a **common cross-fleet visual cue**, integrated with metaheuristic traffic control. The present study addresses this gap by combining a YOLO26m light-bar presence detector with a GA–SGD optimized signal controller through a priority override layer.

B. SYNTHESIS AND IDENTIFICATION OF RESEARCH GAP

Despite significant advancements in emergency vehicle preemption and vision-based traffic monitoring, we observe that a critical gap remains in linking global traffic optimization with dedicated detection of the siren/light-bar as the primary object of interest. We further note that existing vision-based approaches predominantly focus on recognizing the vehicle class (e.g., ambulance or police car), even though vehicle appearance varies considerably across manufacturers and countries. At the same time, acoustic-based systems attempt to infer siren activation but do not provide reliable spatial localization, thereby limiting their effectiveness for precise and real-time traffic control applications. Neither approach directly targets the physical presence of the siren unit, which constitutes the only consistent identifier across heterogeneous emergency fleets.

Furthermore, current frameworks rarely combine hybrid optimization algorithms such as GA–SGD with a non-intrusive visual supervisory layer triggered by this shared visual cue. Most priority mechanisms rely on simplistic green-extension logic that can disrupt globally optimized traffic rhythms. This research addresses these limitations by introducing a YOLO26m module focused on detecting the existence of the roof-mounted siren/light-bar, independent of flashing state or vehicle appearance. Integrating this detector as a supervisory priority layer for a pre-optimized GA–SGD controller provides a novel solution that preserves systemic efficiency while enabling timely emergency-priority intervention within the scope of the proposed framework. The objective is not to replace traffic optimization but to provide a supervisory safety-priority layer that preserves the engineered signal plan during non-emergency periods.

III. INTEGRATED SYSTEM ARCHITECTURE

The proposed system is designed as a dual-stream supervisory control architecture that combines long-horizon traffic optimization with a low-latency emergency override mechanism. The architecture separates efficiency-oriented control from event-driven safety intervention, allowing emergency-priority decisions to be handled independently from the adaptive optimization layer. Within the scope of the present study, this design supports fast supervisory response while preserving the role of adaptive optimization during normal traffic conditions.

The architecture consists of three coordinated components:

- a steady-state optimization layer,
- a priority override layer, and
- a real-time priority arbitration mechanism.

A. STEADY-STATE OPTIMIZATION LAYER (GA–SGD)

Under normal traffic conditions, intersection control is governed by a hybrid Genetic Algorithm–Stochastic Gradient Descent (GA–SGD) optimization engine established in prior work. This layer receives continuous vehicle count, lane occupancy, and flow density data from a standard vision-based traffic detection camera.

The GA component performs global exploration of feasible phase timing configurations, while the SGD component refines solutions locally to minimize average delay and queue length. The optimized signal parameters are transmitted directly to the Traffic Light Controller (TLC) without reliance on external simulation environments. In this pre-deployment framework, GA–SGD functions as the Standard Operating Procedure (SOP) for the intersection, continuously adapting signal timings to evolving traffic conditions in real time.

B. PRIORITY OVERRIDE LAYER (YOLO26M SIREN MODULE)

Operating in parallel to the optimization loop, the YOLO26m Light-Bar Detection Module functions as an independent, high-priority environmental monitor. The model is trained to detect the visual structure of roof-mounted siren/light-bars, independent of whether the lights are flashing. This object-centric formulation ensures robustness across different emergency vehicle designs, colors, and regional fleet variations.

The module is decoupled from traffic-flow optimization to allow independent, high-frequency inference without computational contention. Average inference latency of approximately 11–14 ms per frame enables low-latency monitoring of approaching vehicles based on the detected presence of the siren unit. In the present framework, the light-bar is treated as a practical visual cue for supervisory emergency-priority triggering, although real-world deployment would benefit from additional contextual verification, such as flashing-state analysis or multi-sensor confirmation.

C. REAL-TIME SUPERVISORY PRIORITY LAYER LOGIC

The interaction between the optimization and override layers is managed by a Priority Arbitrator, which governs control transfer based on siren detection confidence and temporal confirmation. The logic flow is defined as follows:

- **Passive Monitoring:** The YOLO26m module continuously evaluates incoming frames. As long as no siren/light-bar object is detected with sufficient confidence, control of the TLC remains with the GA–SGD optimization layer.
- **Priority Trigger:** When the presence of a siren/light-bar is detected with confidence exceeding the pre-defined threshold, the Priority Arbitrator issues a high-priority control signal to the traffic light controller (TLC). A detection confidence threshold of 0.5 is used together with a temporal confirmation rule requiring detection in N consecutive frames ($N = 3$) to reduce transient false triggers. These parameters are consistent with practical YOLO deployment strategies in safety-oriented vision applications, where moderate confidence thresholds combined with short temporal validation windows help balance sensitivity and false-alarm mitigation.
- **Override Execution:** Upon receiving the supervisory priority signal, the TLC enters Emergency Mode, safely terminating the current phase after a mandatory yellow clearance interval and forcing a green phase along the approach path of the emergency vehicle. During this interval, GA–SGD computations are paused.
- **Resumption and Recovery:** Once the emergency vehicle exits the detection zone, the Arbitrator deactivates the supervisory priority signal. Control is returned to the GA–SGD module, which resumes optimization and adjusts subsequent cycles to dissipate any residual congestion introduced by the priority override.

The supervisory priority process is summarized in Algorithm 1. The algorithm formalizes the interaction between YOLO26m-based light-bar detection, temporal confirmation, emergency priority activation, and recovery to the GA–SGD optimized signal plan.

The overall architecture of the proposed framework, shown in Fig. 1, is intended to provide a low-latency, emergency-priority response while preserving long-term traffic efficiency during normal operation.

IV. MODULE METHODOLOGY: DETECTION MODULE AND SYSTEM-LEVEL VALIDATION

This section describes the data engineering process, architectural selection, training protocol, and system-level validation procedure used to integrate the siren detection module into the overarching Genetic Algorithm–Stochastic Gradient Descent (GA–SGD) traffic control framework. The proposed detector is designed to recognize the presence of emergency light-bars mounted on vehicles, independent of their activation state. The dataset includes images where the light-bar is both ON

Algorithm 1 Emergency-Aware Supervisory Priority Control

Input: Optimized GA–SGD signal plan Π^* , video frame F_t , confidence threshold τ , confirmation window N , emergency distance thresholds ρ

Output: Traffic light command C_t and preemption event log

```

1: Set controller mode  $\leftarrow$  Normal
2: Set detection counter  $c \leftarrow 0$ 
3: while the traffic-control system is active do
4:   Acquire current video frame  $F_t$ 
5:   Run YOLO26m light-bar detection on  $F_t$ 
6:   Obtain emergency vehicle position and next traffic light from SUMO/TraCI
7:   if light-bar confidence  $\geq \tau$  then
8:      $c \leftarrow c + 1$ 
9:   else
10:     $c \leftarrow 0$ 
11:   end if
12:
13:   if  $c \geq N$  and vehicle is inside the detection zone then
14:     Estimate distance to junction, signal state, and remaining green time
15:     if priority condition is satisfied and mode = Normal then
16:       Store current traffic-signal phase
17:       Apply yellow clearance interval
18:       Force or extend green phase on the emergency vehicle approach
19:       mode  $\leftarrow$  Emergency
20:       Log preemption start
21:     end if
22:   end if
23:
24:   if mode = Emergency then
25:     Maintain emergency green while the vehicle remains in the priority zone
26:     if emergency vehicle clears the controlled area then
27:       Restore the GA–SGD optimized signal plan  $\Pi^*$ 
28:       mode  $\leftarrow$  Normal
29:        $c \leftarrow 0$ 
30:       Log preemption end
31:     end if
32:   else
33:     Apply normal GA–SGD signal command from  $\Pi^*$ 
34:   end if
35: end while

```

and OFF. Consequently, the learning objective is formulated as a single-class object detection problem targeting the existence of the light-bar structure, rather than flashing state classification. The baseline traffic signal timing is governed by the GA–SGD optimization model previously validated in [1]. The current study does not modify the optimization logic itself but introduces a supervisory visual detection module capable of issuing temporary priority overrides during emergency events. To evaluate the integrated framework beyond detector-level accuracy, a SUMO-based system-level validation was additionally conducted using a modeled representation of the Souissi roundabout in Rabat, Morocco.

A. DATA ACQUISITION AND CURATION

To promote generalization across geographical regions and environmental conditions, we employed a multi-source data

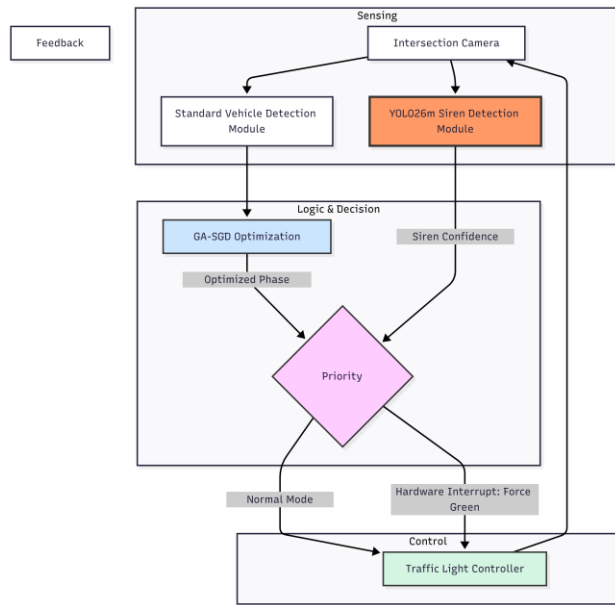


FIGURE 1. System architecture diagram.

acquisition strategy. The primary dataset was collected from three open-source repositories on Roboflow Universe, each focused on emergency vehicle and siren-specific imagery [22], [23], [24]. These sources were supplemented with high-resolution, copyright-compliant images from the public domain to capture a wide variety of siren designs, vehicle types, and intersection scenarios. The final corpus consists of 3,130 images, divided into training, validation, and testing subsets according to the splits provided by the platform.

A rigorous quality control and curation phase was conducted. All images and labels were manually reviewed and corrected using the Roboflow annotation interface to ensure high label fidelity. Unlike conventional vehicle detection tasks, annotations were restricted exclusively to the siren/light-bar region as a physical object, regardless of whether the lights were activated. The dataset intentionally includes both ON and OFF states so that the model learns to detect the existence of the siren unit rather than flashing behavior. Images exhibiting severe blur, extreme occlusion, or ambiguous viewpoints were removed to maintain consistent visual quality.

The detection objective is formulated as a single-class object presence problem targeting the roof-mounted siren/light-bar. The model does not attempt to infer flashing state or classify the entire vehicle. This formulation reflects the fact that the roof-mounted siren/light-bar provides a recurring visual cue across many emergency fleets despite regional design variations. At the same time, the model is intentionally limited to object presence detection and does not infer flashing-state activation or operational status. Consequently, the model does not differentiate ON vs OFF states and does not perform vehicle-type classification.

B. DATASET PARTITIONING

The final curated dataset consisted of 3,130 images, which were divided into three mutually exclusive subsets to enable unbiased training and evaluation. The partitioning strategy prioritized model optimization while preserving sufficient data for validation and final testing:

- **Training Set:** 2,599 images ($\approx 83\%$) used for weight optimization and feature learning.
- **Validation Set:** 266 images ($\approx 8\%$) used to monitor convergence behavior and mitigate overfitting.
- **Test Set:** 265 images ($\approx 8\%$) reserved exclusively for final performance evaluation.

This split ensured statistical independence between training and evaluation stages.

C. ARCHITECTURAL SELECTION: YOLO26M

The YOLO26m (Medium) architecture was selected as the detection backbone because of its suitability for real-time, safety-critical edge deployment. As a recent generation of the YOLO family, YOLO26 introduces end-to-end inference and training improvements that enhance deployment efficiency and small-object detection. These characteristics make YOLO26m a suitable choice for latency-sensitive intelligent transportation applications. Its selection was motivated by the following factors:

- **NMS-Free End-to-End Inference:** YOLO26m eliminates conventional Non-Maximum Suppression (NMS) through a native end-to-end prediction strategy. This design supports stable low-latency inference, which is important for the supervisory emergency-override logic considered in this study.
- **Small-Target-Aware Label Assignment (STAL):** The architecture incorporates specialized label-assignment and loss strategies for small-object detection. This capability is particularly important because the roof-mounted light-bar typically occupies only 3–7% of the frame area at intersection distances, making reliable localization challenging at a fixed input resolution of 640×640 .
- **Accuracy-Latency Trade-off:** The medium-scale variant provides sufficient representational capacity to achieve high precision ($\approx 74\%$), thereby reducing false-positive emergency triggers while maintaining low inference latency (11–14 ms) compatible with embedded intersection hardware.

D. TRAINING SETUP AND CONFIGURATION

The YOLO26m detector was trained in a cloud-based Google Colab environment using a free-tier NVIDIA T4 GPU (16 GB GDDR6). All experiments used pre-trained Ultralytics weights and were configured for a single-class object-detection task targeting the roof-mounted siren/light-bar. Training settings were selected to ensure stable convergence, reproducibility, and real-time deployment feasibility.

Key training parameters:

- **Task:** object detection
- **Input resolution / image size:** 640×640 pixels
- **Batch size:** 16
- **Epochs:** 100
- **Optimizer:** SGD (default Ultralytics configuration)
- **Initial learning rate (lr0):** 0.01
- **Final learning rate fraction (lrf):** 0.01
- **Momentum:** 0.937
- **Weight decay:** 0.0005
- **Early stopping patience:** 25 epochs
- **Data augmentation:** Mosaic and MixUp, applied to improve robustness to varying illumination, occlusion, and background complexity typical of urban intersections

The detector was trained without architectural modification, and all runs were executed under identical settings, varying only the random seed to assess robustness against stochastic initialization effects.

Reproducibility protocol:

- **Default configuration** (framework seed)
- **Seed:** 42
- **Seed:** 123

Average inference latency during validation:

- **Default run:** 11.45 ms
- **Seed 42:** 13.16 ms
- **Seed 123:** 13.89 ms

These results correspond to approximately 72–87 FPS on an NVIDIA T4 GPU, indicating real-time-rate inference under the evaluated experimental conditions. Final performance is reported as mean $\pm$ standard deviation in the Experimental Results section.

E. SUMO-BASED SYSTEM-LEVEL VALIDATION

To validate the operational behavior of the proposed integrated framework beyond detector-level performance, a system-level simulation study was conducted in the SUMO environment using a modeled representation of the Souissi roundabout in Rabat, Morocco. The purpose of this validation was to assess whether the light-bar detection module, when integrated as a supervisory layer over the pre-optimized GA–SGD traffic controller, could improve emergency vehicle passage while preserving overall traffic efficiency.

1) STUDY AREA SELECTION

The Souissi roundabout in Rabat, Morocco, was selected for system-level validation because it constitutes a strategically relevant urban traffic node in the vicinity of several major hospital facilities. The roundabout lies within approximately 0.65–3.6 km of CHU Souissi, Hôpital Universitaire International Mohammed VI - Rabat, CHU Ibn Sina, and Hôpital Cheikh Zayd, corresponding to estimated travel times of roughly 2–7 minutes. Owing to this proximity, the site provides a realistic context for ambulance movement and emergency-priority evaluation, making it suitable for testing

the operational effect of the proposed vision-triggered supervisory control framework.

2) COMPARED CONTROL STRATEGIES

Three traffic control strategies were evaluated under identical traffic demand conditions:

- **Fixed-time control:** representing a conventional non-adaptive baseline without emergency prioritization;
- **GA–SGD:** representing the previously optimized adaptive traffic signal controller operating under normal traffic conditions but without emergency-priority logic;
- **Proposed:** representing the integrated framework in which the GA–SGD controller was augmented by the vision-triggered emergency override mechanism.

This comparative design makes it possible to distinguish the benefit of adaptive optimization alone from the additional contribution of the proposed supervisory emergency layer.

3) SIMULATION ENVIRONMENT AND EMERGENCY SCENARIO DESIGN

The validation was implemented through a dedicated Python-SUMO experiment pipeline. The emergency vehicle was modeled as a dedicated ambulance route traversing the Souissi roundabout network. To avoid relying on a single arbitrary test case, the simulation was performed under multiple traffic and emergency timing conditions.

In particular:

- passenger traffic demand was scaled using three candidate levels: 10, 12, and 14;
- ambulance departure time was varied across seven candidate values: 310 s, 315 s, 320 s, 325 s, 330 s, 335 s, and 340 s.

This stress-testing design allowed the proposed framework to be evaluated under different temporal interactions between the emergency vehicle, queue formation, and signal state.

4) PREEMPTION LOGIC WITHIN THE PROPOSED CONTROLLER

Within the proposed strategy, emergency prioritization was implemented as an event-driven override policy operating on top of the optimized GA–SGD controller. The simulation logic used a predefined emergency profile to determine when temporary priority should be activated based on emergency vehicle proximity and traffic state.

The selected validation profile used the following settings:

- detection distance: 450 m;
- decision distance: 380 m;
- priority distance: 280 m;
- *green-hold* and *green-extension* parameters to support emergency corridor clearance.

When the emergency condition was satisfied, the controller temporarily overrode normal adaptive timing and granted priority passage. After the emergency vehicle cleared the

controlled area, traffic-signal operation was returned to the GA-SGD optimized cycle.

5) EVALUATION METRICS AND PAPER-SCENARIO SELECTION

Simulation outputs were collected from SUMO trip-information files and event logs. The following metrics were used to evaluate performance:

- *ambulance waiting time;*
- *ambulance travel time;*
- *total waiting time for all vehicles;*
- *average waiting time for all vehicles;*
- *average travel time for all vehicles;*
- *number of preemption activations in the proposed scenario.*

Based on the full demand-departure sweep, a representative paper scenario was selected for detailed comparison. The final paper-level analysis was reported for the configuration corresponding to demand scale 12 and ambulance departure time 330 s, as this case provided a balanced stress condition in which emergency delay remained observable under GA-SGD while the proposed framework produced measurable improvements at both the emergency-vehicle level and the network level.

Overall, this SUMO-based methodology provides system-level validation of the integrated framework by linking detector-triggered supervisory control to measurable traffic-performance outcomes in a realistic roundabout setting.

The modeled Souissi roundabout network used for SUMO validation is illustrated in Fig. 2.



FIGURE 2. SUMO-based modeled representation of the Souissi roundabout in Rabat, Morocco, used for system-level validation of the proposed framework. The figure illustrates the multi-approach geometry and the complexity of the simulated traffic environment.

V. EXPERIMENTAL RESULTS

A. QUANTITATIVE DETECTION PERFORMANCE

The performance of the proposed YOLO26m-based siren detection module was evaluated using standard object detection metrics, including Precision, Recall, mAP50, and mAP50-95, as shown in Table 1. To assess robustness against stochastic initialization effects, three independent training runs were conducted under identical hyperparameter and dataset configurations, varying only the random *seed*.

TABLE 1. Detection performance across independent training runs.

Run	Precision	Recall	mAP50	mAP50-95
Default	0.7058	0.6964	0.7085	0.3306
Seed 42	0.7610	0.6920	0.7488	0.3465
Seed 123	0.7239	0.6944	0.7170	0.3391

To evaluate reproducibility, performance metrics were aggregated across the three independent runs. The model achieved:

- **Mean Precision:** 0.730 ± 0.028
- **Mean Recall:** 0.694 ± 0.002
- **Mean mAP50:** 0.725 ± 0.021
- **Mean mAP50-95:** 0.339 ± 0.008

The results demonstrate consistent detection performance across independent training executions. The model achieves reliable localization accuracy despite variations in emergency vehicle design, camera viewpoint, and lighting conditions.

The mAP50-95 metric, which penalizes localization errors across multiple IoU thresholds, confirms stable object boundary learning for the roof-mounted siren/light-bar.

All qualitative visualizations correspond to the best-performing model (*Seed* = 42), while quantitative results are reported as mean $\pm$ standard deviation across three independent runs.

The training and validation loss curves are shown in Fig. 3.

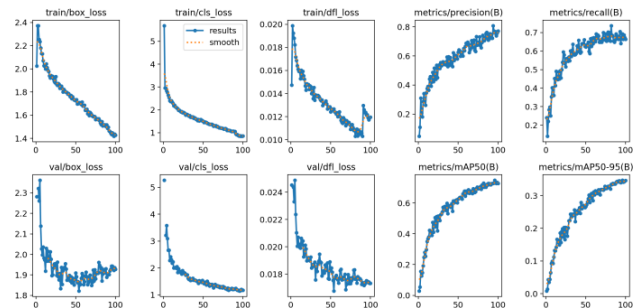


FIGURE 3. Training and validation loss convergence curves across epochs for the YOLO26m siren detection model.

B. STABILITY AND REPRODUCIBILITY ANALYSIS

To evaluate training robustness, the standard deviation across three runs was computed. The observed variance is minimal, particularly for recall (± 0.002) and mAP50-95 (± 0.008), indicating stable convergence behavior.

This low variability confirms that performance is not dependent on favorable initialization conditions and that the model consistently learns the siren/light-bar representation.

Such reproducibility is critical for safety-oriented traffic control systems, where detection reliability must not be sensitive to stochastic training effects.

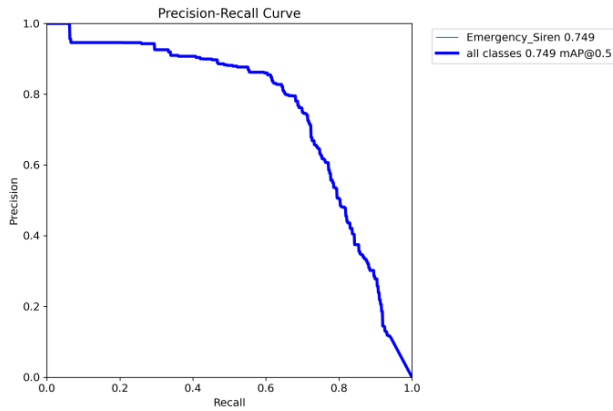


FIGURE 4. Precision–Recall curve demonstrating detection stability across confidence thresholds.

The results therefore indicate stable performance across the three independent training runs under the present experimental conditions.

C. PRECISION–RECALL CHARACTERISTICS

The Precision–Recall (PR) curves further illustrate detection stability. All runs exhibit smooth convergence behavior with no significant oscillatory instability, confirming proper training dynamics.

The PR curve demonstrates balanced sensitivity and precision across confidence thresholds, supporting the selected detection confidence threshold of 0.5 combined with temporal confirmation.

The precision–recall behavior is presented in Fig. 4.

D. CONFUSION MATRIX ANALYSIS

The normalized confusion matrix, shown in Fig. 5, indicates strong discrimination capability for the single-class detection task. False positive and false negative rates remain within acceptable bounds given the challenging nature of rooftop small-object detection under variable lighting.

The results confirm that the detector focuses on the siren/light-bar region rather than the vehicle body, validating the object-level formulation of the detection objective.

E. QUALITATIVE DETECTION RESULTS

Visual inspection of the validation predictions, as shown in Fig. 6, confirms that the model successfully localizes roof-mounted sirens across:

- Different emergency vehicle types
- Variable intersection camera angles
- Daytime and partial occlusion conditions
- Both ON and OFF flashing states

Importantly, the detector does not depend on flashing behavior and correctly identifies the physical presence of the siren unit even in inactive states, supporting the design rationale of using the light-bar as a robust object-level visual cue within the proposed framework.

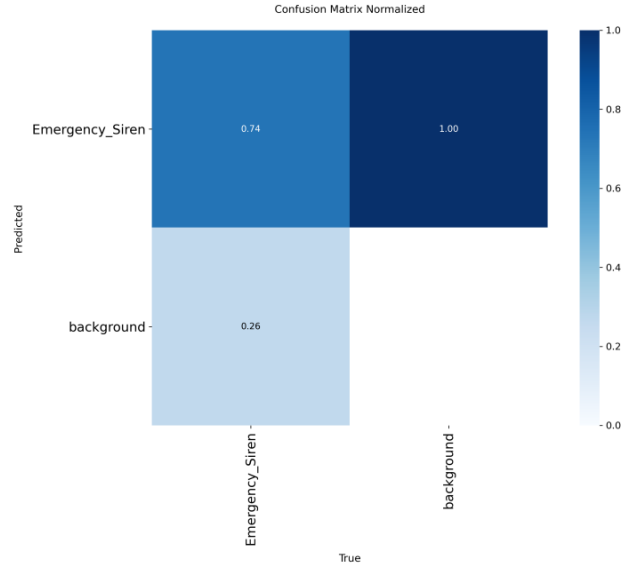


FIGURE 5. Normalized confusion matrix for single-class siren/light-bar detection.



FIGURE 6. Representative validation detections of the proposed YOLO26m model illustrating successful localization of roof-mounted emergency light-bars across diverse environmental conditions and scale variations.

F. REAL-TIME FEASIBILITY

Inference latency measured during validation was:

- 11.45 ms (Default run)
- 13.16 ms (Seed 42)
- 13.89 ms (Seed 123)

This corresponds to approximately 72–87 FPS on an NVIDIA T4 GPU, indicating real-time-rate inference under the evaluated experimental conditions.

The computational efficiency supports the integration of the detection module as a supervisory priority layer without introducing perceptible system latency.

Beyond detector-level evaluation, a system-level simulation study was conducted to assess the operational impact of integrating the proposed light-bar detection module with the GA-SGD traffic controller. The following subsections report the SUMO-based validation results for the Souissi roundabout case study introduced in *Section IV-E*.

G. SUMO-BASED COMPARATIVE PERFORMANCE IN THE SELECTED PAPER SCENARIO

To evaluate the operational impact of the proposed framework, the SUMO-based case study described in *Section IV-E* was used to compare three traffic control strategies under the selected paper scenario: Fixed-time control, GA-SGD, and the proposed GA-SGD + vision-triggered priority override. According to the experiment pipeline, the final paper scenario corresponded to demand scale **12** and ambulance departure time **330 s**, which was selected from the calibrated demand-departure sweep for detailed comparison.

The results show that the proposed integrated framework achieved the best overall emergency response performance. Most importantly, the proposed controller produced the lowest ambulance travel time, reaching **56 s**, compared with **70 s** under GA-SGD and **64 s** under Fixed-time control. This corresponds to a reduction of approximately **20.0%** relative to GA-SGD and **12.5%** relative to Fixed-time control. Ambulance waiting time was also reduced to **15 s**, compared with **21 s** under GA-SGD and **22 s** under Fixed-time control. Although waiting time is not the strongest standalone metric, this still represents a meaningful **28.6%** reduction relative to GA-SGD in a safety-critical scenario.

At the network level, the proposed strategy also yielded the lowest total waiting time for all vehicles, with a value of **19,887 s**, compared with **22,827 s** for GA-SGD and **33,719 s** for Fixed-time control. This result is particularly important because it indicates that emergency prioritization was not achieved at the expense of general traffic performance. On the contrary, the proposed controller reduced total network waiting time by approximately **12.9%** relative to GA-SGD and by approximately **41.0%** relative to Fixed-time control. The same trend is observed for average traffic indicators, where the proposed method achieved the lowest average waiting time for all vehicles (**14.154 s**) and the lowest average travel time for all vehicles (**76.398 s**), compared with **16.247 s** and **79.618 s** under GA-SGD and **23.999 s** and **89.653 s** under Fixed-time control.

In the selected paper scenario, the proposed strategy triggered one preemption event, confirming that the observed gains were associated with an explicit supervisory intervention rather than a passive variation in traffic conditions. Overall, these findings demonstrate that the proposed framework improves emergency passage while maintaining, and in this case even improving, overall traffic efficiency.

The SUMO comparison results are summarized in Table 2.

TABLE 2. SUMO-based comparison of controller performance in the selected paper scenario.

Scenario	Ambu Wait (s)	Ambu Travel (s)	Total Wait (s)	Avg Wait All (s)	Avg Travel All (s)	Preempt
Fixed-time	22	64	33,719	23.999	89.653	0
GA-SGD	21	70	22,827	16.247	79.618	0
Proposed	15	56	19,887	14.154	76.398	1

H. SYSTEM-LEVEL INTERPRETATION OF THE INTEGRATED FRAMEWORK

The SUMO results indicate that the potential contribution of the proposed method extends beyond emergency light-bar detection. Within the proposed supervisory architecture, the detector serves as a decision trigger intended to modify traffic signal behavior in favor of emergency passage. This distinction is important: the model is not proposed merely as a perception module, but as a control-enabling component within an integrated intelligent transportation framework.

From a system perspective, the strongest result is the simultaneous improvement in **ambulance travel time** and **global traffic performance**. In many emergency-priority strategies, local gains for the emergency vehicle are obtained at the cost of increased congestion for the surrounding traffic stream. In contrast, the present results show that the proposed architecture reduced ambulance travel time while also producing the lowest total waiting time across the network. This indicates that the vision-triggered override was sufficiently targeted and temporally limited to support emergency passage without destabilizing the broader traffic-control process.

These findings therefore support the design rationale introduced in Sections III and IV: the GA-SGD controller remains responsible for long-horizon efficiency under normal conditions, while the vision-based emergency layer intervenes only when safety-critical conditions are detected. The SUMO case study demonstrates that this interaction can be translated into measurable traffic benefits at the system level.

I. SENSITIVITY ACROSS EMERGENCY DEPARTURE CONDITIONS

To avoid basing the evaluation on a single arbitrary emergency event, the simulation study included a departure-time sweep in which the ambulance entry time was varied across multiple candidate values while maintaining the selected demand level. This sensitivity analysis showed that the effect of emergency prioritization depends on the temporal interaction between the ambulance, existing queue formation, and the active signal state at the moment of detection. As expected, not all departure conditions produced identical gains.

Nevertheless, the selected paper scenario at demand scale 12 and departure time 330 s provided the most representative and balanced comparison for detailed reporting. In this case, emergency delay remained observable under GA-SGD, while the proposed strategy produced improvements in

ambulance waiting time, ambulance travel time, and total network waiting time simultaneously. This makes the selected scenario particularly suitable for demonstrating the intended role of the proposed framework as an emergency enhancement layer applied on top of an already optimized adaptive controller.

The sensitivity analysis also reinforces an important interpretation of the framework: the proposed system should not be viewed as a permanently dominant alternative to adaptive control, but rather as a context-sensitive supervisory mechanism that becomes active only when emergency passage requires immediate signal arbitration. This behavior is consistent with the intended safety-priority role of the light-bar detection module.

VI. DISCUSSION

A. DETECTION RELIABILITY AND SYSTEM-LEVEL RELEVANCE

The proposed YOLO26m-based siren/light-bar detector achieved stable performance across independent training runs, with low variability in recall and mAP50–95, indicating reliable convergence for the target object. Although the task remains challenging because the light-bar is a small rooftop object, the results show that the detector learned a consistent representation suitable for supervisory emergency-cue detection. This interpretation is reinforced by the SUMO-based validation, where the perception module also functioned as a control-enabling component within the integrated traffic-management framework.

B. SUPERVISORY CONTROL VALUE AND PRACTICAL FEASIBILITY

A key strength of the proposed framework is its non-intrusive supervisory design. By focusing on the roof-mounted siren/light-bar rather than full-vehicle classification, the method reduces dependence on vehicle appearance and regional fleet variability. Integrated as a supervisory layer over the GA–SGD controller, the detector allows emergency-priority intervention without permanently altering the underlying adaptive optimization logic. In addition, inference speeds of approximately 11.45–13.89 ms per frame confirm computational feasibility for intersection-level deployment studies, while temporal confirmation helps reduce transient false triggers.

C. POSITIONING RELATIVE TO EXISTING EMERGENCY-PRIORITY APPROACHES

Compared with conventional emergency vehicle preemption approaches, the proposed framework offers a practical balance between robustness and deployability. IoT- and V2I-based methods require dedicated infrastructure, while acoustic siren-detection approaches are sensitive to urban noise and limited spatial localization. Vision-based vehicle-classification methods provide localization, but they depend on recognizing entire vehicle categories that may vary across

TABLE 3. Controlled architectural comparison on identical dataset (seed = 42).

Model	Precision	Recall	mAP50	mAP50–95
YOLOv8m	0.720	0.685	0.691	0.325
YOLO26m	0.761	0.692	0.749	0.346

regions and fleets. In contrast, the present siren-centric strategy preserves the spatial advantages of vision while reducing dependence on vehicle-body appearance, making it suitable for heterogeneous urban contexts.

D. RELIABILITY, SCENARIO DEPENDENCE, AND MODEL CHOICE

The proposed framework should be interpreted as a safety-oriented enhancement to adaptive control rather than as a universal replacement for optimization-based traffic management. The low standard deviation across training runs supports the reliability of the perception layer, while the SUMO study across multiple demand scales and ambulance departure times supports the plausibility of the control layer in simulation. At the same time, the benefit of emergency prioritization remains scenario-dependent, which justifies interpreting the method as a context-sensitive supervisory mechanism rather than as a permanently dominant strategy. The detector also targets light-bar presence rather than flashing-state activation; therefore, within the present study it should be viewed as a visual emergency-cue module in a pre-deployment framework rather than as a complete real-world authorization mechanism for signal preemption.

The architectural comparison further supports the choice of YOLO26m for the present task. Under identical training conditions, YOLO26m outperformed YOLOv8m in precision, recall, mAP50, and mAP50–95, indicating better localization quality for compact targets such as roof-mounted light-bars. A contextual comparison with related YOLO-based emergency-detection studies also suggests that object-level light-bar detection can achieve competitive performance while reducing dependence on vehicle geometry, livery, and manufacturer, although such comparisons should be interpreted cautiously because of differences in datasets and labeling protocols.

The architectural comparison with YOLOv8m is reported in Table 3.

YOLO26m demonstrated improved performance for the specific task of small-scale light-bar detection under identical experimental conditions.

The contextual comparison with related work is shown in Table 4.

VII. LIMITATIONS

Although the study includes detector-level evaluation and SUMO-based system validation, several limitations remain. First, the dataset is still moderate in size, and the proposed task remains challenging because light-bar detection

TABLE 4. Contextual comparison with related YOLO-based emergency detection studies.

Study	Target	Precision	Recall	mAP50
Satyanarayana <i>et al.</i> (2025) [18]	Ambulance (vehicle level)	0.762	0.631	0.69
Proposed (YOLO26m - best run)	Light-bar (object-level)	0.761	0.692	0.749

Note: mAP50–95 was not reported in Satyanarayana *et al.* (2025). Results are not directly comparable due to differences in dataset composition, annotation protocols, and detection formulation.

is inherently a small-object detection problem under variable viewpoint, scale, and illumination conditions. Second, the current validation is simulation-based and relies on a simplified closed-loop setup with controlled emergency scenarios. Full roadside validation was beyond the scope of the present study because testing emergency-priority traffic interventions in real urban conditions requires formal authorization, institutional coordination, and safety clearance from public authorities. Third, the study is limited to a single modeled roundabout, which restricts generalizability to larger and more complex traffic networks. Finally, the current model does not explicitly classify flashing-state activation or operational status; therefore, the proposed detector should be interpreted as a visual emergency-cue module rather than a complete deployment-ready authorization system. In addition, large-scale deployment may require further optimization for lightweight edge hardware.

VIII. CONCLUSION

This study proposed an emergency-aware adaptive traffic signal control framework that combines a vision-based siren/light-bar detection module with a globally optimized GA–SGD traffic controller. Unlike conventional approaches based on acoustic sensing or full-vehicle classification, the proposed method detects the roof-mounted siren/light-bar as an object-level visual cue for emergency-aware supervisory control. The YOLO26m detector, trained on a curated dataset of 3,130 annotated images, showed stable performance across multiple runs, confirming the feasibility of this single-class object-presence formulation.

The main contribution of the present work is the system-level validation of the integrated framework. The detection module operates as a supervisory layer over the GA–SGD controller, preserving normal traffic efficiency while enabling temporary emergency-priority intervention. SUMO-based experiments on a modeled Souissi roundabout in Rabat showed that the proposed strategy achieved the lowest ambulance travel time and the lowest total waiting time for all vehicles, while also reducing ambulance waiting time compared with GA–SGD. These results support the practical value of combining emergency prioritization with adaptive traffic optimization in a pre-deployment simulation setting.

Although the study remains limited to simulation-based validation, it demonstrates the promise of integrating object-level emergency detection with adaptive traffic control. Future work should extend the framework through broader multi-scenario validation, tighter closed-loop integration, flashing-state or context-aware verification, and eventual real-intersection deployment on edge-computing platforms.

CONFLICT OF INTEREST

The authors declare no conflicts of interest.

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